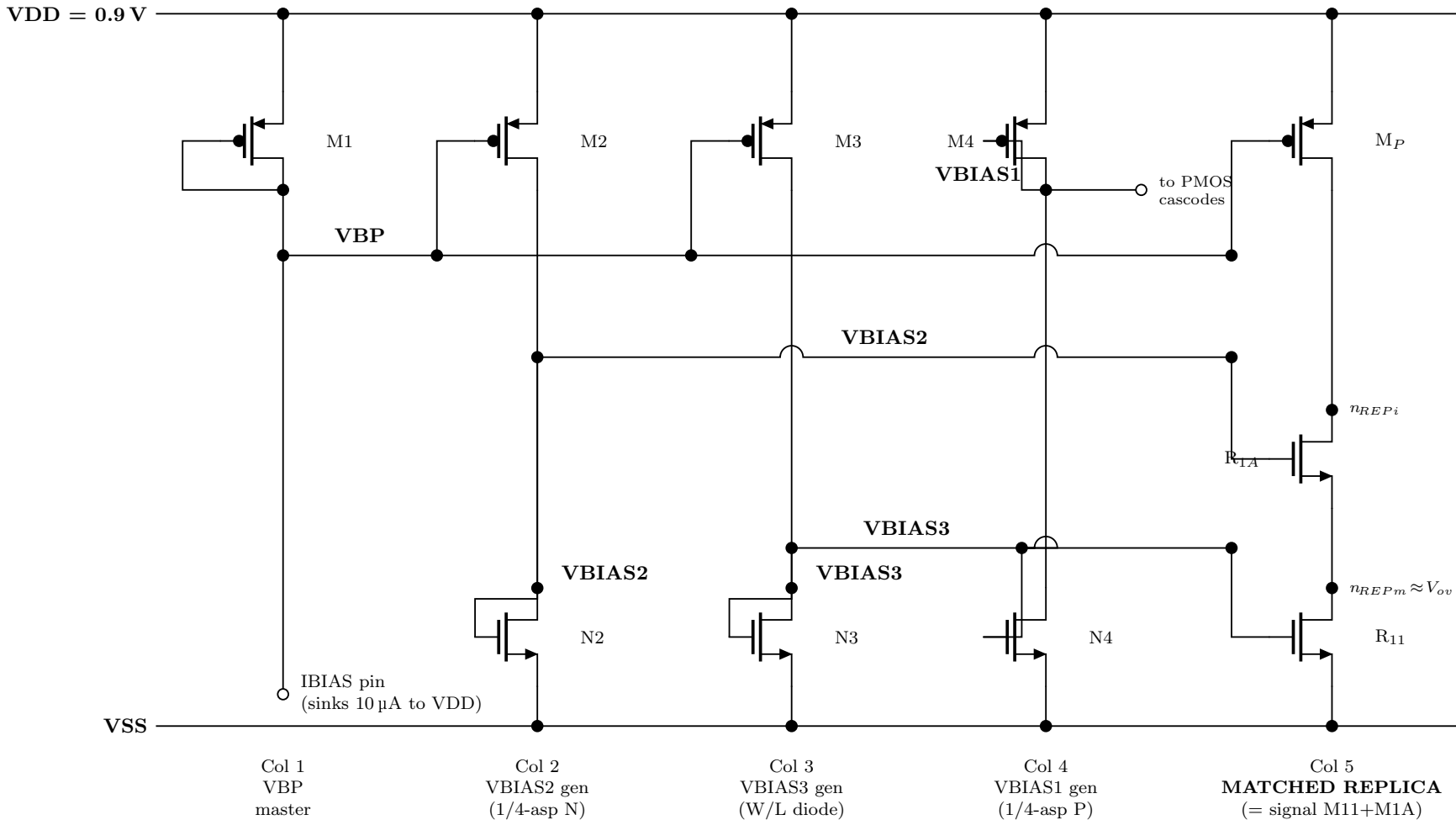


Sooch wide-swing cascode bias — VDD = 0.9 V (matched- V_{ds})



Device	Sizing (sky130_fd_pr LVT)	Role	m	I [μ A]
M1 = XMP_REF	L=1, W=16, nf=4	master PMOS diode (sets VBP)	4	10
M2 = XMP_B2	L=1, W=16, nf=4	PMOS mirror (g=VBP), col 2	4	10
M3 = XMP_VB3	L=1, W=16, nf=4	PMOS mirror (g=VBP), col 3	4	10
M4 = XMP_B1	L=4, W=16, nf=4	1/4-asp PMOS diode (sets VBIAS1)	4	10
MP = XMP_REP	L=1, W=16, nf=4	PMOS mirror for matched replica (g=VBP, 2 $\times$)	8	20
N2 = XMN_B2	L=2, W=4, nf=2	1/4-asp NMOS diode (VBIAS2 = $V_T + 2V_{ov}$)	4	10
N3 = XMN_REF	L=0.5, W=4, nf=2	NMOS diode (VBIAS3 = $V_T + V_{ov}$)	4	10
N4 = XMN_B1	L=0.5, W=4, nf=2	NMOS sink (g=VBIAS3) for col 4	4	10
R1A = XMN_R1A	L=4, W=32, nf=4	replica cascode (g=VBIAS2) – unit cell of signal M1A	1	20
R11 = XMN_R11	L=0.5, W=4, nf=2	replica mirror (g=VBIAS3) – unit cell of signal M11	2	20

corner	TT	FF	SS	FS / SF
Matched- V_{ds} verification (5/5 PVT corners, measured 2026-06-02):				
$V_{ds}(M_{11})$ signal = nbL	73	72	71	81 / 61 mV
$V_{ds}(R_{11})$ replica = nREPM	68	65	71	72 / 64 mV
Δ	5	7	0.1	9 / 3 mV

Why this is the right Sooch topology at 0.9 V: (1) At sub-1 V, stacking a third diode-tied device in a bias column creates a high-impedance floating source node that picks the wrong DC basin in TT/FF/FS. We avoid this by keeping every *gate-voltage generator* (cols 1–4) two devices tall. (2) The wide-swing matching is recovered by a separate *cascoded NMOS replica* (col 5) whose gates are externally driven by the already-stable VBIAS2 / VBIAS3 rails – no diode-tie loops, single stable DC basin. (3) The replica current is scaled (20 μ A = one signal-half sink) so the per-unit current density matches the signal: R_{11} (m=2 @ 20 μ A) and M_{11} (m=4 @ 40 μ A) both run at 10 μ A/unit, identical V_{ov} per unit - i_c cascode action pins $V_{ds}(R_{11}) = V_{ds}(M_{11})$, cancelling λ -mismatch in the VBIAS3 voltage mirror. Verified: VOUTP locked to $V_{CM}=0.3$ V, $A_0 = 43$ –50 dB, GBW = 1.25–1.86 MHz, PM = 80–83°, IDD = 45–50 μ A across all 5 PVT corners.